EXPERIMENT 12

Implementing Security Features and Governance / Administration Practices in Zoho Creator

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Aim

To implement and explore the security, access control, data-sharing and governance features of a Zoho Creator application.

Introduction

Security and governance in low-code applications deal with controlling who can use an application, what each person is permitted to do, and how the data and resources of the application are administered. Because low-code platforms make it easy to build applications that store business data, controlling access to that data is an important part of application design.

Zoho Creator provides built-in mechanisms for this purpose. User permissions define the operations that a user may perform on the components of an application. Role-based access places users in roles that are arranged in a hierarchy, which influences the records that they can see. Data sharing rules extend the visibility of records from one role or user to another with a defined level of access. Administrative controls allow the application to be configured and managed. Together, they help ensure that users receive only the access needed for their responsibilities and that application data is protected from unauthorised use.

Application Used

The practical was carried out in Zoho Creator using the Student Admission Enquiry Management application. The application captures prospective-student enquiries through the Admission_Enquiry_Form, which stores personal details, contact information, academic preferences and consent. Because these records contain personal data of applicants, the application served as a suitable practical environment for exploring the security and governance features described in this record. [If your Zoho Creator account was on an expired trial and some features required an upgrade, state it here; otherwise delete this highlighted note.]

Security and Governance Architecture

Figure 1 is a conceptual supporting diagram of the security and governance structure of the Student Admission Enquiry Management application. It is not a screenshot of the Zoho Creator interface.

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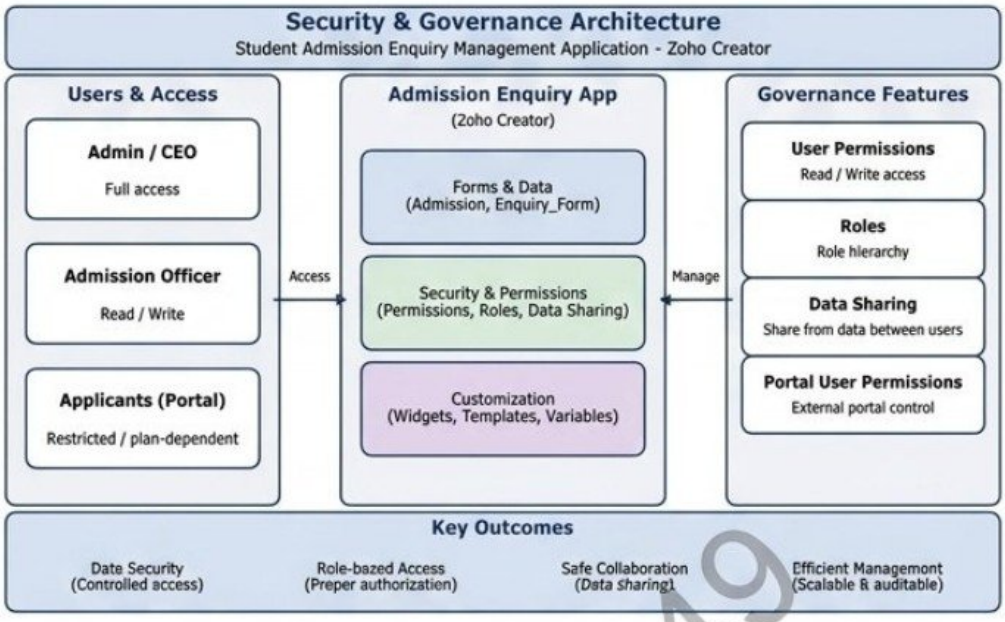


Figure 1: Security & Governance Architecture (Conceptual Diagram)

The diagram shows users (such as the CEO / administrator, admission officers with Read / Write access and external portal users such as applicants) accessing the application, which contains forms and data, security and permissions (user permissions, roles and data sharing) and customisation items. The governance features on the right (user permissions, roles, data sharing and portal user permissions) manage and control access to the application. The key outcomes are data security through controlled access, role-based access, safe collaboration through data sharing, and efficient management.

User Permissions

The User Permissions section is opened from Settings in the application editor. It contains the tabs Permissions, Roles and Data Sharing, and is used to add and manage the permissions and roles of users and to define the visibility of records for each user. The Permissions lab hiss the permission sets available in the application and provides on Add Permission option. The application provided two predefined permissions:

Name	Description (as displayed)
Read	This profile will have read permission for all components
Write	This profile will have write permission for all components

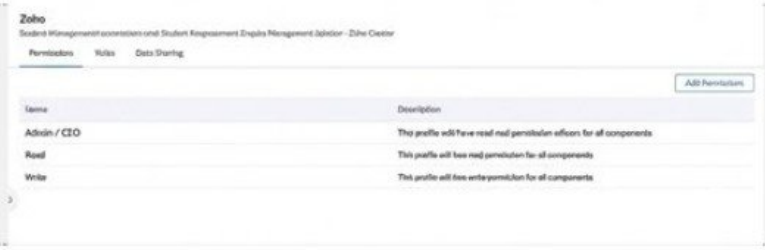


Figure 2: User Permissions in the Student Admission Enquiry Management application

Read permission allows a user to view the components and records of the application without changing them, whereas Write permission allows the user to make changes, such as adding or modifying enquiry records.

Role-Based Access Control

The Roles tab shows the role hierarchy of the application. In the application, the default role CEO is displayed, and the options Expand All, Collapse All, Manage Role Hierarchy and Add Role are available.



Figure 3: Roles and Role Hierarchy in the Student Admission Enquiry Management application

The Add Role option was used to configure a custom role with the following details:

Field	Value
Role Name	Admission Officer
Reports To	CEO
Description	Manages student admission enquiries
Share Data with Peers	Not selected
Status	[Saved / Not saved – an upgrade was required: choose what applies to your account]

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Figure 4: Add Role Panel– Admission Officer Role Configuration

The Admission Officer role was configured with the required name, reporting hierarchy and description, so that front-office staff report to the CEO role in the hierarchy. [Complete this paragraph according to your result: e.g. 'The role was saved and now appears under CEO in the hierarchy.' OR 'Saving the custom role required an upgrade because the trial had expired, so the CEO role remains the only role.']

Data Sharing

Data sharing rules control the visibility of the records of a form. A rule specifies the form, the role or user from which records are shared, the role or user to which they are shared, and the level of access given. In this way, records can be made visible to additional roles or users beyond the default visibility of the role hierarchy. A data sharing rule was configured with the following details:

Parameter	Value
Rule Name	Admission Enquiry Data Sharing
Form	Admission Enquiry Form
Shared From	CEO (role)
Shared To	MADHUMITHA P (user)
Access Level	Read/Write

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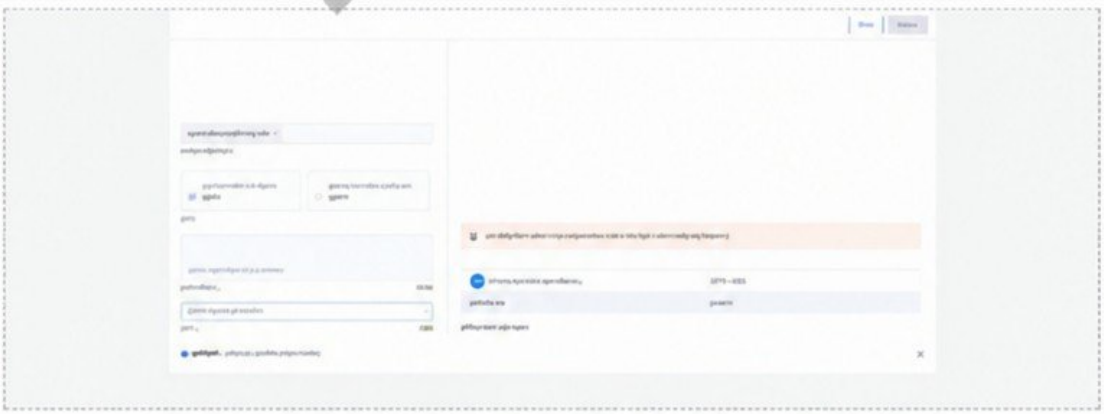


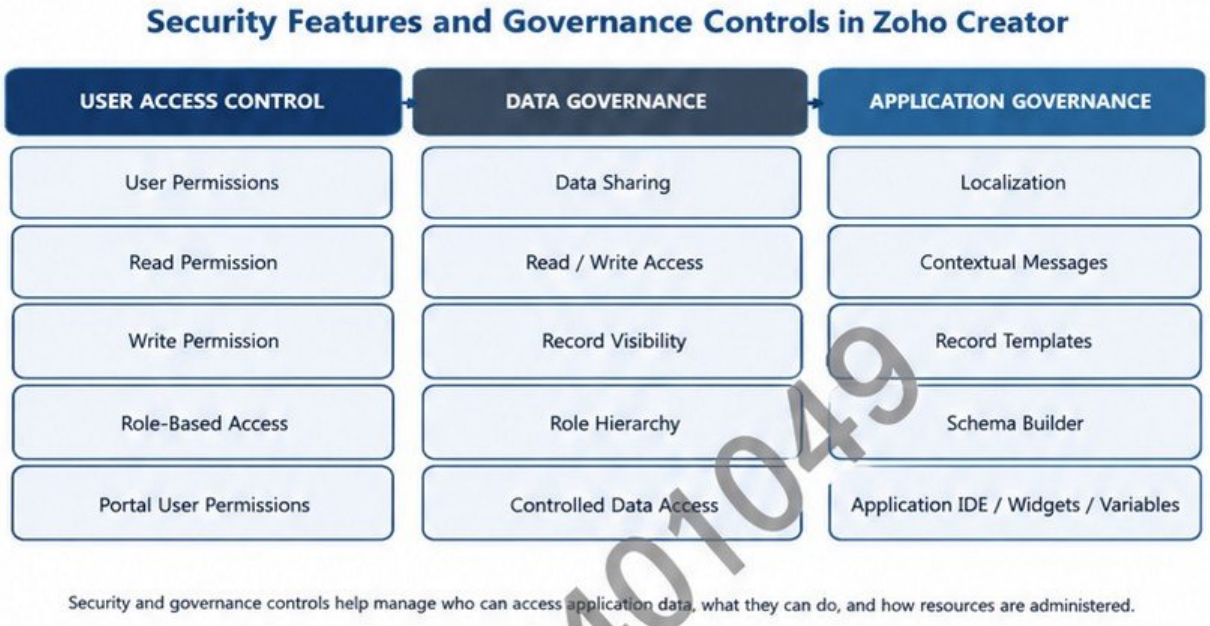
Figure 5: Data Sharing rule configured for the Admission Enquiry Form

The saved rule appears in the Data Sharing list with the columns Name, Form, Shared From, Shared To and Level of access. With Read/Write access, the user DHIVYA R can both view and modify the shared records

of the Admission Enquiry Form that belong to the CEO role. The level of access determines how much a recipient can do with the shared records, so that visibility is extended in a controlled manner.

Security Features and Governance Controls

Figure 6 is a conceptual academic diagram that groups the security and governance controls of Zoho Creator. It is not a screenshot, and it is not proof that every listed feature was configured.



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Figure 6: Security Features and Governance Controls in Zoho Creator (Conceptual Diagram)

Control	Explanation
User access control	User permissions (Read and Write), role-based access and portal user permissions determine who can use the application and what they can do.
Data governance	Data sharing, read / write access, record visibility and role hierarchy govern how application data is accessed.
Application governance	Settings such as localization, contextual messages, record templates, schema builder, application IDE, widgets and variables are used to administer and customise the application.
Role hierarchy	Arranges roles in a reporting structure that influences which records a role can see.
Data sharing	Extends the visibility of records from one role or user to another with a chosen access level.
Record visibility	Decides which records are visible to which users, based on roles and sharing rules.
Administrative controls	Allow an administrator to manage permissions, roles and application settings.

In this practical, user permissions and roles were examined, and a data sharing rule was configured. Mention any items restricted by your account, e.g. Portal User Permissions or saving the custom role, if

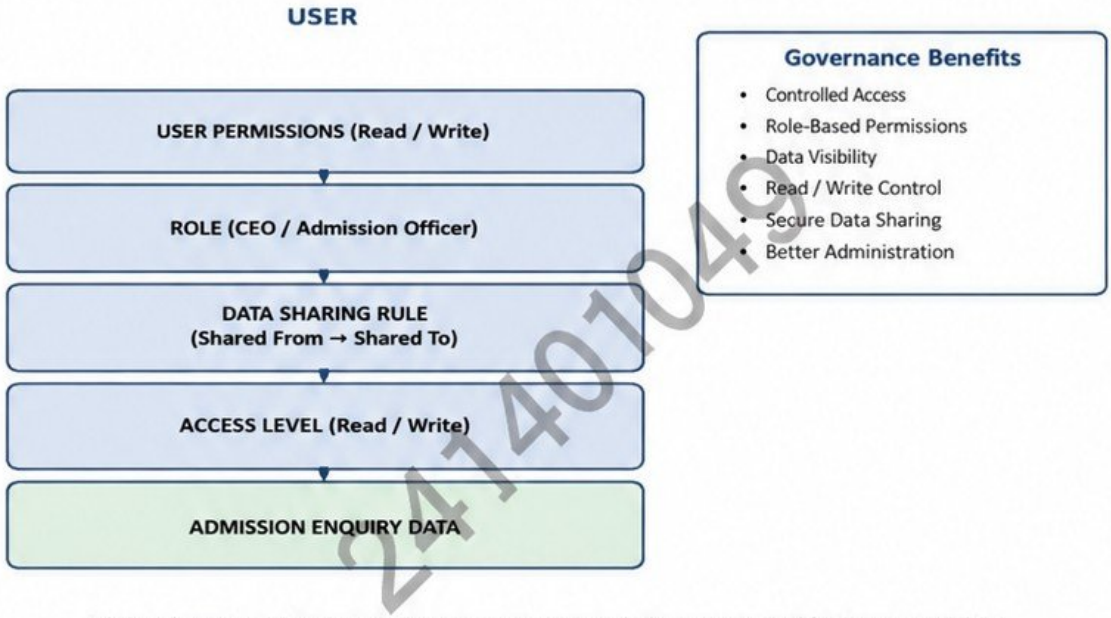
applicable.] The other items in the diagram, such as localization, schema builder and widgets, are available in the Settings menu of the application, but they were not configured in this experiment.

Portal User Permissions

Portal User Permissions is listed beside User Permissions in the Settings menu and is used to control the access of external portal users. For this application, such users would be prospective students who submit enquiries through a portal rather than through the public form link. [State whether it was only accessed for examination or configured, and whether an upgrade was required.]

Role-Based Access and Data Sharing Flow

Figure 7 is a conceptual diagram of the security flow: User → User Permissions → Role → Data Sharing Rule → Access Level → Admission Enquiry Data. It represents how access is controlled in principle and is not evidence that each step was configured.



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Figure 7: Role-Based Access and Data Sharing Flow (Conceptual Diagram)

- A user is first given permissions (Read / Write).
- The user belongs to a role, such as CEO or Admission Officer, which determines the position in the hierarchy.
- A data sharing rule specifies the records shared from one role or user to another.
- The access level (Read / Write) defines what the recipient can do, and this controls access to the Admission Enquiry data.

Governance Observations

- Predefined Read and Write permissions were available in the application.
- Roles could be accessed, and a custom role configuration could be prepared.
- [Add your observation on saving the custom role.]

- Data Sharing was configured with Read/Write access.
- [Add your observation on Portal User Permissions.]
- The practical demonstrated how access to sensitive applicant data can be controlled through permissions, roles and sharing rules in a low-code platform.

Conclusion

Zoho Creator provides built-in mechanisms for managing users, roles, permissions and data visibility. In this experiment, the predefined Read and Write permissions were examined, the role hierarchy with the default CEO role was viewed, and a data sharing rule was configured to share the records of the Admission Enquiry Form from the CEO role to the user DHIVYA R with Read/Write access.

Some advanced governance capabilities may depend on the subscription plan. [Add a sentence on any plan restrictions you faced, or delete this note.] The experiment shows the access-control tools that Zoho Creator offers. Controlling who can access application data, and at what level, remains an essential part of building and managing business applications, especially for applications that hold personal details of applicants.

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